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Repel-kill-release: A self-polishing antifouling coating with synergistic effects of fluorescent antifouling and contact bacteriostasis

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ABSTRACT

Marine antifouling coatings offer an effective solution to the issue of marine fouling; however, maintaining the delicate balance between long-term efficacy and environmental sustainability presents significant challenges. Herein, we have developed a novel self-polishing surface that combines UV fluorescence-responsive antifouling and contact antifouling. This is achieved using a stabilized acrylate polymer as the backbone, with rosin and rosin-derived quaternary ammonium salts (MPA-N⁺) suspended from the side chains. It exhibits excellent “offensive capabilities” for repelling and killing fouling organisms, as well as “defensive capabilities” for surface release. Both rosin and MPA-N⁺ exhibit a fluorescent response to the UV in seawater with high transmittance, repelling marine organisms such as algae. MPA-N⁺ endows acrylate polymers with strong “offensive” capabilities, allowing the coating to effectively penetrate the defenses of bacteria and algae, leading to significant inactivation. Additionally, the release function and surface renewal capabilities of the acrylate polymer facilitate the continuous regeneration of functional surfaces, ensuring prolonged durability and environmentally friendly antifouling properties. The resin polymers demonstrate excellent antifouling effects owing to the synergistic combination of fluorescent antifouling, contact biocide, and surface self-renewal. The coated surface maintains antifouling efficacy for over 240 days in real marine environments. This study introduces promising strategies for achieving a balance between long-term effectiveness and environmental sustainability in marine antifouling.

1. Introduction

Marine biofouling is a widespread phenomenon affecting nearly all underwater surfaces, including ships [1], bridges [2], underwater pipelines [3], and various marine instruments [4], resulting in significant negative impacts on ocean exploration [5–7]. Antifouling coatings represent one of the most effective and feasible countermeasures against the detrimental phenomenon of marine biofouling adhesion [8,9]. These coatings utilize two primary antifouling strategies: the “defensive strategy,” which resists the adhesion of biofouling to the surface [10], and the “offensive strategy,” which eliminates fouling organisms near the surface.

For “defence strategies”, self-polished surfaces based on hydrolysis or degradation have been proposed [11–13]. They generate hydrophilic surfaces through the hydrolysis of polymer side chains and facilitate surface self-renewal in the presence of seawater [14]. The continuous

self-renewal of the surface is effective for releasing organic and inorganic substances that adhere to the coating, which is important to resist the adhesion of marine organisms fouling. However, self-polishing coatings alone are not inherently resistant to biofouling adhesion and cannot effectively address the complexities of the marine environment through surface-released “defense strategies” alone [15]. Therefore, integrating “offensive strategies” with polymers that possess surface release or self-renewal capabilities represents a promising approach for achieving durable marine antifouling solutions.

Antimicrobial compounds, including isothiazolinones [16], amphoteric ions [8], natural compounds [17], and quaternary ammonium salts [18], are essential to the “offensive strategy” and have been utilized to develop coated surfaces with outstanding microbicidal capabilities. Among these, quaternary ammonium salts (QAS) are particularly favored for constructing antimicrobial surfaces due to their low toxicity, broad-spectrum efficacy, high efficiency, and stable bactericidal

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properties [19,20]. Zhang et al. successfully grafted amphoteric sulfo-propyl betaine (ISB) and quaternary ammonium salt (QAS) onto the surface of cotton fabrics using a reactive isocyanate group (NCO), thereby imparting long-term antimicrobial activity to the fabrics [21]. Kinetic antimicrobial experiments demonstrated that the antimicrobial rate against the two tested bacteria increased by 99.8 % within the first 0.5 h, with complete bacterial eradication achieved after 2 h of contact. The surface demonstrated excellent antibacterial adhesion properties. The injection of QAS into a slippery liquid-infused porous surface (SLIPS) facilitated contact sterilization and reduced microbial adhesion [22].

Among various quaternary compounds, rosin-derived quaternary ammonium salt (MPA-N⁺) not only exhibits broad-spectrum antimicrobial properties but also displays a UV fluorescence response. This functionality can be attributed to its raw material, rosin—a natural product abundant in nature—known for its antimicrobial activity, resistance to protein adsorption, and a UV fluorescent response with a maximum emission wavelength around 430 nm [23]. Notably, seawater is highly transparent to ultraviolet radiation [24]. Studies have shown that in the eastern Mediterranean at mid-latitudes, the ability of seawater to attenuate UV light is low. UV intensity at 375 nm decreases by only 5 per cent for every 1 m increase in water depth; For visible light (blue light) 465 nm, the corresponding value is also only 3 %. In contrast, in highly transparent seawater, UV radiation can be as deep as 20 m, which allows the photochemical processes of algae to proceed normally [25]. UV fluorescent-responsive materials combat marine biofouling at the initial stage by minimizing diatom deposition on coated surfaces [26]. The mechanism by which light-responsive materials enhance antifouling in coatings is based on the relationship between cellular physiological activity and light, influencing the spatial interference between chloroplasts and hold-fast structures [27]. With increased light, diatoms prioritize photosynthesis over settling [28], which effectively prevents the attachment of algae. Thus, fluorescence response emerges as a potential method for inhibiting fouling by primary organisms like diatoms [29].

Herein, we have developed a novel self-polishing surface that combines UV fluorescence-responsive antifouling with contact antifouling, utilizing rosin and rosin-derived quaternary ammonium salts (MPA-N⁺) to provide effective repellent and antimicrobial “offensive capabilities” alongside surface-release “defensive capabilities”. Acrylate prepolymers were synthesized through straightforward free radical polymerization reactions using acrylic and acrylate monomers. The prepolymer was then combined with rosin, MPA-N⁺, and zinc oxide to prepare acrylate resin polymers featuring rosin and MPA-N⁺ side chains via an acid-base reaction. This resin polymer demonstrates synergistic effects of fluorescent antifouling and contact bactericidal properties, resulting in excellent marine antifouling performance. Additionally, the acrylate polymers can undergo hydrolysis, providing excellent surface renewal properties. This process not only releases Zn²⁺ (~0.1 mg/g) and natural rosin to assist in sterilization and inhibition of algal growth but also removes fouling materials adhering to the coating surface, enabling the regeneration of functional surfaces with strong biocidal activity. We investigated the properties of the coatings, including surface morphology, self-polishing ability, transparency, flexibility, resistance to protein adhesion and biofouling, as well as their applicability in real marine environments. Our goal is to develop an antifouling coating that can be directly applied in marine settings, featuring efficient and environmentally friendly properties.

2. Experimental section

2.1. Materials

Ethyl acrylate (EA), butyl acrylate (BA), acrylic acid (AA), methacrylate (MAA) were purchased from Datang Chemical Group Co. Xylene, n-butanol, and butyl acetate were purchased from Datang Chemical

Group Co. Dodecanethiol was purchased from Sinopharm Chemical Reagent Co. Azodiisobutyronitrile (AIBN) was purchased from Tianjin Damao Chemical Reagent Factory. Zinc oxide (ZnO) were purchased from Sinopharm Chemical Reagent Co. Rosin was purchased from Zhiyuan Chemical Products Co. Materials for the preparation of quaternary ammonium salts of maleic acid: Maleic anhydride, N,N-dimethylethylenediamine and ethylene bromide were purchased from Shanghai Aladdin Biochemical Technology Co; Acetic acid, p-toluenesulfonic acid, ethanol, tetrahydrofuran were purchased from Sinopharm Chemical Reagent Co. The materials used in the test: 2216E agar and 2216E liquid medium were purchased from Hai Bo Biotechnology Co. The seawater was obtained from Qingdao sea.

2.2. Synthesis of MPA-N⁺

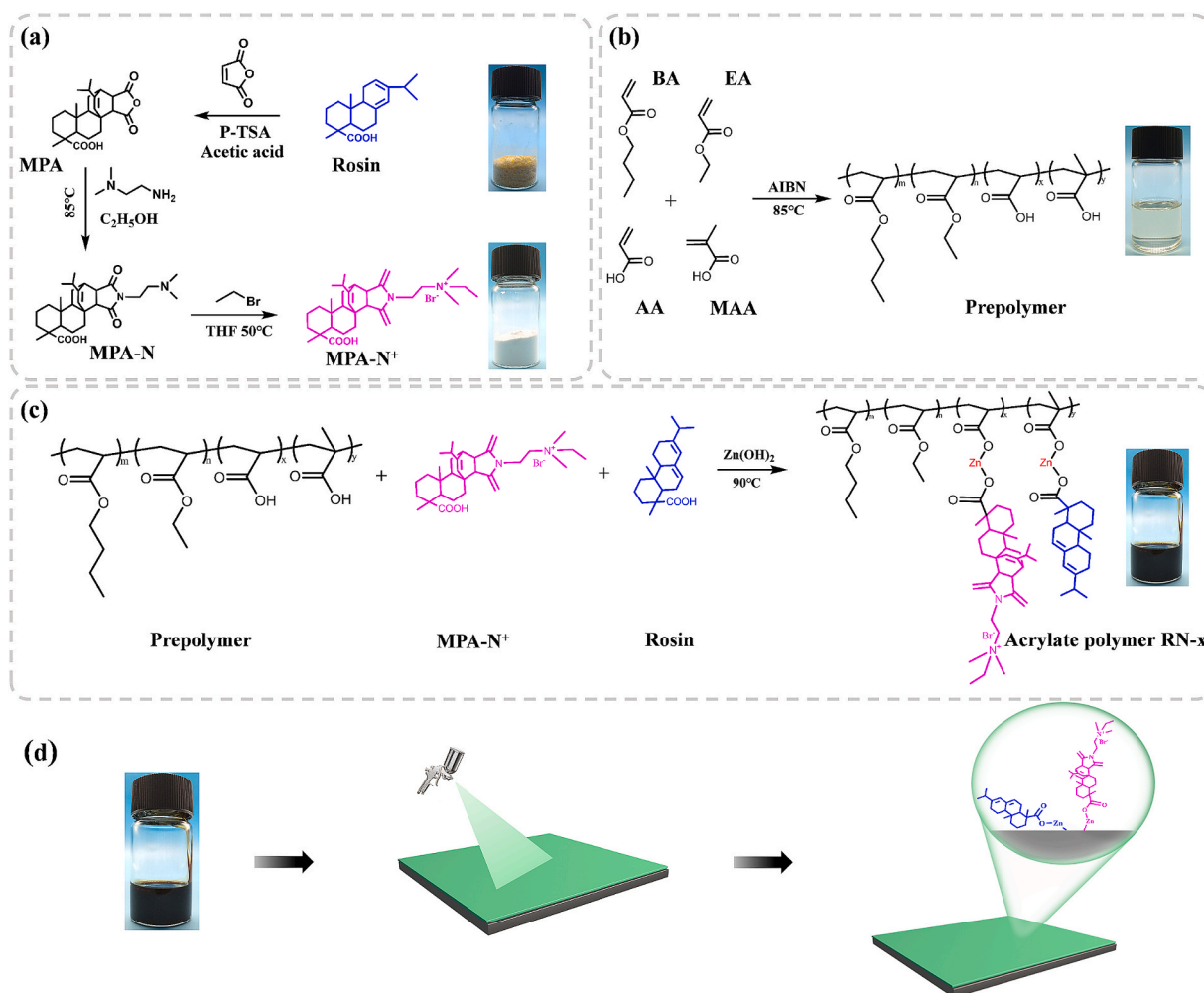
As shown in Scheme 1a, the preparation of MPA-N⁺ followed previously reported methods [26]. Rosin acid (100.0 g, 0.28 mol) was heated under a nitrogen atmosphere via condensation reflux at 180 °C for 3 h. Subsequently, the reaction temperature was reduced to 120 °C, and maleic anhydride (27.5 g, 0.28 mol), p-toluenesulfonic acid (P-TSA) (5.2 g, 0.028 mol, catalyst) and acetic acid (100.0 mL) as solvent were added. The reaction was condensed and refluxed at 120 °C for 24 h, after which it was cooled and allowed to stand for 2 h. The crude product was washed and recrystallized with acetic acid to obtain maleic acid (MPA). Next, 10 g of MPA was dissolved in 200 mL of ethanol, and N,N-dimethylethylenediamine (2.8 mL, 0.025 mol) was added dropwise. The mixture was reacted at 85 °C for 3 h, and upon standing and cooling, crystals of the product precipitated. These crystals were washed with ethanol to yield MPA-N. Subsequently, 10 g of MPA-N (0.02 mol) was dissolved in 200 mL of tetrahydrofuran, and ethidium bromide (31 mL, 0.02 mol) was added. The reaction was conducted at 50 °C for 24 h, after which the crude product was precipitated, filtered, and washed three times with tetrahydrofuran to obtain the final product, MPA-N⁺.

2.3. Synthesis of acrylate prepolymer

In a beaker, 23.9 g of butyl acrylate (BA), 18.7 g of ethyl acrylate (EA), 4.0 g of methacrylic acid (MAA), 3.4 g of acrylic acid (AA), 0.32 g of AIBN were mixed thoroughly. The mixed solution was divided into 10 portions, with one portion placed in a 250 mL three-necked flask. Sequentially, xylene (40 g), n-butanol (10 g), and n-dodecanethiol (0.2 g) were added, and the mixture was heated to 85 °C. Subsequently, one portion of the mixed solution was added at 15-minute intervals, and finally, AIBN (0.08 g) was added to maintain the reaction for 4 h, yielding an acrylate copolymer with a solid content of 50 %. The synthetic illustration of prepolymer was shown in Scheme 1b.

2.4. Synthesis of MPA-N⁺ modified acrylate polymers (RN-x)

As shown in Scheme 1c, MPA-N⁺-modified acrylate resin copolymers were labeled as RN-x, where x = 0, 5, 10, 15, 20, 25, or 30 indicates the amount of MPA-N⁺ used (i.e., 0, 2.33, 4.67, 7.00, 9.33, 11.66, and 14.00 mmol, respectively). The preparation involved taking 50 g of acrylate prepolymer (46.67 mmol of COOH), the desired amounts of MPA-N⁺ (n = 0, 2.33, 4.67, 7.00, 9.33, 11.66, or 14.00 mmol), and natural rosin (46.67-n mmol), and adding them to a three-necked flask. ZnO (3.8 g, 46.67 mmol) was added, and the required solvent mixture (xylene: n-butanol: butyl acetate = 27.50: 6.25: 66.25 w %) was added to ensure that the solid content of the acrylate copolymer was 50 %. Table S1 (Supplementary Material) provides the feed molar ratios of the synthesized materials. Subsequently, the polymer was coated onto ABS plates for further testing; specific steps in the fabrication of the polymer sample plates can be found in the Supplementary Information.



Scheme 1. Synthesis of (a) MPA-N⁺, (b) acrylate prepolymer, (c) acrylic polymer RN-x.

2.5. Characterization

2.5.1. Scanning electron microscopy (SEM) and energy dispersive X-ray spectroscopy (EDX)

The surface and cross-sectional morphology of the coatings were examined using scanning electron microscopy (SEM), specifically with a Zeiss Gemini 300, applying a voltage of 10 kV during observation. Prior to testing, the samples were coated with gold to enhance conductivity. The surface elements of the coatings were observed using energy dispersive EDX.

2.5.2. ¹H nuclear magnetic resonance (¹H NMR) spectra

The ¹H NMR spectra of MPA-N⁺ and RN-x were recorded using a Bruker AVANCE III 600 NMR spectrometer, with DMSO serving as the solvent.

2.5.3. Thermogravimetric analysis (TG)

A thermogravimetric analyzer (STA449C, NETZSCH, Germany) was employed for the thermodynamic analysis of the coatings. Tests were conducted in a nitrogen atmosphere, covering temperatures from 20 °C to 800 °C, with a heating rate of 10 °C/min. The differential thermogravimetric analysis (DTG) curves of the samples were also evaluated.

2.5.4. X-ray photoelectron spectroscopy (XPS)

The elemental composition and chemical state of the polymer surface were analyzed using a Thermo Scientific K-Alpha X-ray photoelectron spectrometer.

2.5.5. Confocal laser scanning microscope (CLSM) observation of surfaces

The surface morphology and roughness of the samples before and after the experiments were observed and recorded using an OLYMPUS OLS 4000 confocal laser scanning microscope. The experiments were divided into static immersion and dynamic seawater simulation experiments, with the experimental setup for the dynamic simulation illustrated in Fig. S1 (Supplementary Material).

2.5.6. Water contact angle (WCA)

The WCA of the samples before and after the experiment was measured using a KRÜSS DSA 100 contact angle automated tester, with a dosing volume of 2 μL. The results were averaged from three parallel samples.

2.5.7. Coating adhesion test

The adhesive strength of RN-x was measured using an AGS-J electronic universal testing machine according to ASTM D4541. The prepared RN-x polymer coatings were applied to test iron columns with a radius of 1 cm, maintaining a thickness of 80 ± 20 μm, and were dried at 40 °C for 3 days. Epoxy AB adhesive was then applied to another test iron post. Subsequently, the two coated surfaces were brought into contact and dried in an oven at 40 °C for an additional 3 days. The adhesive strength for each sample was calculated by averaging five experimental measurements.

2.5.8. Fluorescence emission spectroscopy test

After the resin coating was dried, its fluorescence emission spectrum

was measured using a fluorescence spectrophotometer. The excitation and emission slit widths were set to 2.5 nm and 5.0 nm, respectively. The excitation wavelength (EX) was 370.0 nm, the emission wavelength (EM) was 440.0 nm, and the scanning voltage was 700 V

2.5.9. Transparency and flexibility tests

Each coating (0.1 g/cm²) was applied to a PET film with a thickness of about 50 μm. The transparency of the coating is observed before and after immersion in seawater. The flexibility of the coating is tested by bending

2.6. Self-polishing performance test

Self-polishing capability is a crucial aspect of the RN-x coating's ability to continually renew its surface and function offensively. The self-polishing ability of the polymer coatings was assessed by measuring the mass loss of the coatings, following previously reported methods [30]. ABS plates (8 × 10 cm²/10 × 26 cm²) coated with polymerized samples were tested in a dynamic seawater simulator after recording their initial mass. Each test cycle lasted 7 days; After that, the samples were removed, cleaned with water, dried and placed in a 45 °C oven at for 3 days before measuring their final mass. After completing the test, the next cycle of experiments was conducted, lasting 35 days on the sample plates. The self-polishing ability was evaluated based on the rate of mass loss.

The rate of mass loss (M) is calculated by the following equation (1):

$$M = (M_0 - M_i)/S \quad (1)$$

where M_0 is the original mass of the sample ABS plate without experiments; M_i is the mass of the ABS plate after i cycles; S represents the area of the plate.

Furthermore, to assess the potential environmental impact of the coating, we quantitatively analyzed the release of Zn from the RN-20 coating with the fastest self-polishing rate in seawater using inductively coupled plasma optical emission spectroscopy (ICP-OES). Specifically, 1 g of the coating polymer was uniformly applied to a 1 × 1 cm² glass substrate and immersed in 50 mL of seawater for 10 days. After immersion, seawater samples were collected and analyzed. The measurements were performed using a Thermo Fisher Scientific iCAP 6000 Series ICP-OES (USA), with a nebulizer flow rate of 0.78 L/min and RF power set to 1500 W.

2.7. Anti-fouling performance test

2.7.1. Antibacterial test

Antimicrobial experiments were conducted using *Escherichia coli* and marine bacteria (*Pseudomonas aeruginosa*). The bacteria were first activated for 24 h in an incubator at 37 °C. The culture medium and agar were prepared and autoclaved. The bacteria were then placed in the medium for another 24 h, after which the solution was diluted. The absorbance of the bacterial suspension was adjusted to 0.2–0.3 at 600 nm using a UV absorption photometer. Subsequently, 1 mL of the bacterial solution was added dropwise to the sample plates (5 × 5 cm²) coated with the polymer, covered with PE film, and placed into Petri dishes (d = 10 cm) sealed with sealing film. The dishes were then placed in a shaking incubator at 37 °C for 24 h. Following this, the sample plates were rinsed repeatedly with saline (10 mL) to ensure the complete removal of bacteria, and the rinse solution was collected as a bacterial stock solution. This stock solution was then diluted with saline to create multiple gradients of diluted bacterial solutions. A volume of 100 μL from each diluted solution was spread evenly on the surface of the agar medium, applying it repeatedly until fully absorbed. The solid medium was subsequently placed in a shaking incubator at 37 °C for 24 h. All items used in the antimicrobial experiments were autoclaved, and the entire process was conducted in a sterile ultra-clean bench. Pure ABS

plates served as the blank control group.

The rate of inhibition (N) is calculated by the following equation (2):

$$N = (1 - N_i/N_c) \times 100\% \quad (2)$$

where N_i and N_c are the average number of colonies in the experimental and control samples, respectively.

2.7.2. Anti-protein adhesion test

The initial adhesion of proteins to surfaces marks the onset of marine fouling organism attachment, highlighting the importance of evaluating the resistance of resin coatings to protein adhesion. Firstly, phosphate buffered saline (PBS) at pH 7.2 was prepared and the resin coating was soaked for 2 h.

A 400 mL solution of bovine serum albumin (BSA) at a concentration of 0.05 mg/mL was prepared using PBS as the solvent. The soaked coatings were immersed in 30 mL of this BSA solution for 24 h, with the coated side facing downward [31]. This mixture was diluted by adding 20 mL of PBS. From it, 1 mL was taken and mixed with 10 mL of the Caumas Brilliant Blue working solution. After thorough shaking, absorbance was measured at using UV spectrophotometry (595 nm) [32]. The concentration of BSA was determined from the standard curve (Fig. S2).

The rate of anti-protein adhesion (C) is calculated by the following equation (3):

$$C = C_i/C_0 \times 100\% \quad (3)$$

where C_0 and C_i are the concentration of unused BSA solution and protein not adhering to the coating.

2.7.3. Inhibition of algal growth and resistance to algal attachment tests

The seawater used in this experiment was natural seawater, initially pump-filtered using a microporous filter membrane (pore size = 0.45 μm). Nutrient salts necessary for algal growth were then added to the seawater (the specific preparation method is detailed in [Supplementary Material Table S2](#)). *Chlorella vulgaris*, *Nitzschia closterium* and *Phaeodactylum tricornutum* were selected for the anti-algal experiments. These three algae were inoculated into the prepared seawater containing nutrient salts and cultured in an intelligent artificial climate chamber (20 °C, 70 % humidity, light: dark time ratio of 12 h:12 h) for 7 days, with regular shaking each day to ensure optimal growth. The algal solution was diluted, and its absorbance was measured using a UV absorption photometer, aiming for a reading of 0.1 at 600 nm, corresponding to the desired concentration. The prepared polymer-coated plates (5 × 5 cm) were then placed in a glass Petri dish (d = 10 cm), and 50 mL of a fixed concentration of algal solution was added. Incubation continued for 7 days, during which the absorbance of the algal solution was recorded daily using a UV absorption photometer to assess the inhibitory properties of the polymer on algal growth.

The rate of Algae inhibition (A) is calculated by the following equation (4):

$$A = (1 - A_i/A_c) \times 100\% \quad (4)$$

where A_i and A_c are the algal concentration in the coated samples and in the blank.

The algae adhesion resistance experiments were conducted following a similar procedure, with the key difference being the focus on measuring the concentration of algae adhered to the surface of the sample plates. The coating's surface was thoroughly rinsed multiple times with seawater (10 mL) to ensure complete collection of all algae. The rinse solution was then analyzed to determine the anti-adhesion properties of the polymer coating.

The rate of Algae Adhesion Resistance (F) is calculated by the following equation (5):

$$F = (1 - F_i/F_c) \times 100\% \quad (5)$$

where F_i is the absorbance of algae attached to the surface of the polymer sample and F_c is the absorbance of algae attached to the surface of the control group.

2.7.4. Actual marine antifouling performance testing

The development of antifouling coatings necessitates support from experiments conducted in real marine environments. To more accurately evaluate the antifouling effectiveness of the developed polymer coatings, we conducted marine antifouling experiments in Qingdao Harbor. PVC boards were selected as the experimental substrates, which were prepared through the following steps: The PVC boards were sanded with sandpaper and cleaned with water to remove any debris. After drying, the resin was applied to the PVC sheet. Once the first coat dried, additional coats of 0.1 g/cm^2 were applied successively. The ocean experiment was conducted in the harbor of Qingdao, where the samples were submerged approximately 1 m below the surface. During the observation phase, the coated plates were initially removed and meticulously cleaned, followed by the photography and documentation of their surfaces. Subsequently, the coated plates were returned to their original positions to maintain experimental integrity. The antifouling performance of the coatings was evaluated by examining the appearance of the sample surfaces.

3. Results and discussion

3.1. Characterization of the RN-x coatings

The surface condition and composition of the coatings were analyzed

using SEM, EDS, ^1H NMR, and XPS techniques. Fig. 1 displays the SEM images and elemental distribution of the RN-20 coating. In Fig. 1a, the cross-sectional morphology of the RN-20 coating reveals a thickness of approximately $280 \mu\text{m}$. The uniform smoothness observed in the cross-section indicates that the coating is consistent throughout its depth. Fig. 1b, c illustrate the surface morphology of the RN-20 coating, which also appears smooth.

To assess the elemental distribution on the surface of the RN-20 coating, elemental analysis was conducted using EDS (Fig. 1d–h). The analysis revealed the presence of C and O, as well as the elements Zn, N, and Br on the coated surface. The elemental spectra indicate that the characteristic elements of RN-20—Zn, N, and Br—are uniformly distributed across the surface of the coating. This observation suggests that MPA- N^+ is effectively dispersed within the coating.

The MPA- N^+ and RN-x polymers were elucidated using ^1H NMR spectroscopy. The ^1H NMR spectrum is presented in Fig. 2, where Fig. 2a displays the NMR spectrum of MPA- N^+ . Characteristic peaks include hypomethyl a ($\delta = 5.38$, $-\text{CH}=\text{}$), methylene b ($\delta = 3.64$, $-\text{CH}_2-$), methylene e ($\delta = 3.04$, $-\text{CH}_3$), and hypomethyl g ($\delta = 2.63$, $-\text{CH}=\text{}$), confirming the successful synthesis of MPA- N^+ [19,33]. ^1H NMR comparisons of RN-0 and RN-20 polymers are presented in Fig. 2b (with ^1H NMR spectra of other RN-x polymers available in Fig. S3 of the Supplementary Material). The figure shows that the characteristic peaks a ($\delta = 5.38$, $-\text{CH}=\text{}$) and e ($\delta = 3.03$, $-\text{CH}_3$) from MPA- N^+ are present in the RN-20 polymer. This confirms the successful grafting of MPA- N^+ onto the polymer and the successful synthesis of RN-20.

The TG-DTG curves of the coatings were analyzed, with results displayed in Fig. 3. Both RN-x exhibited similar heat absorption peaks. In the TG curve, only minimal mass loss was observed in region I (between

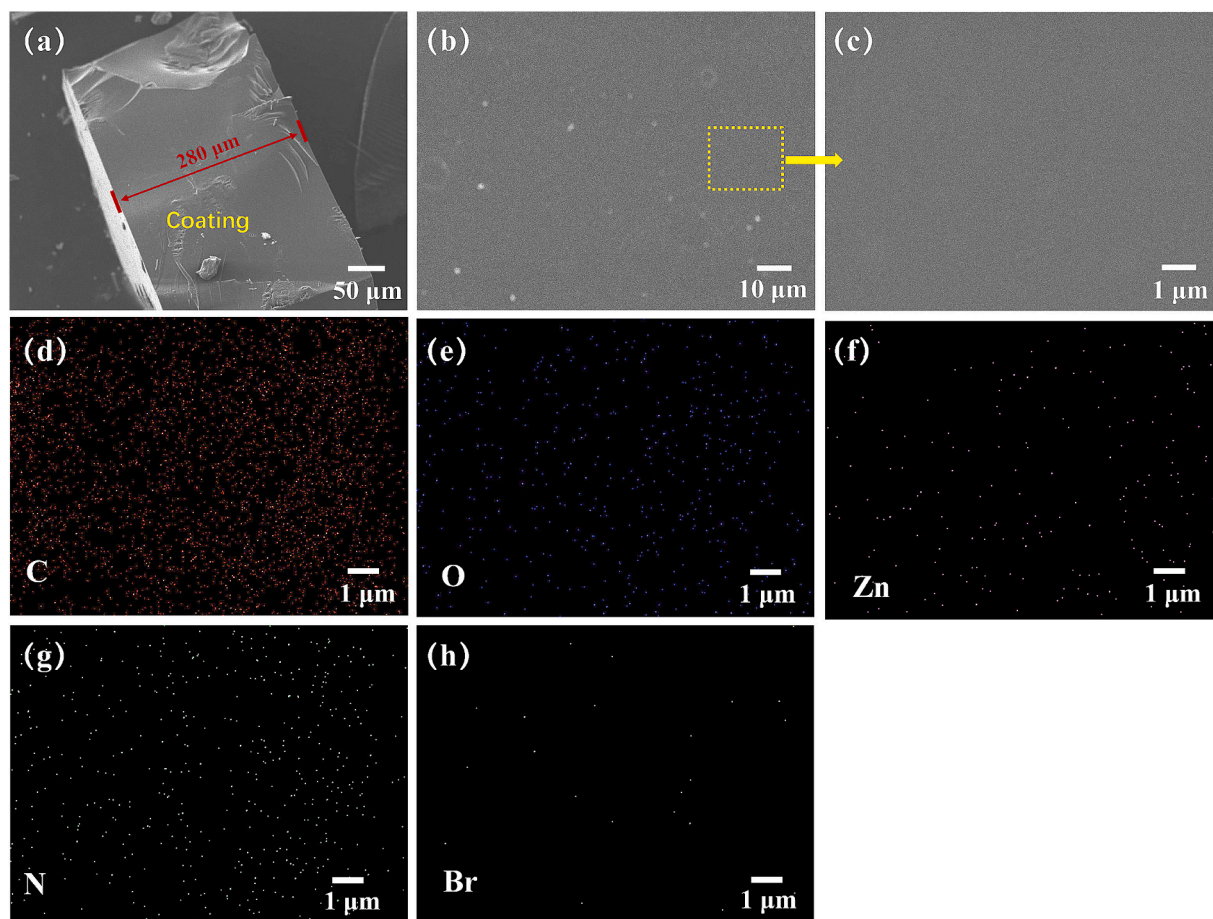


Fig. 1. SEM and EDS mapping of RN-20 coating: (a) Cross-section morphology and (b, c) Surface morphology of the RN-20 (panel (c) is part of panel (b) at high magnification) and (d–h) mapping of element distribution of RN-20 coating.

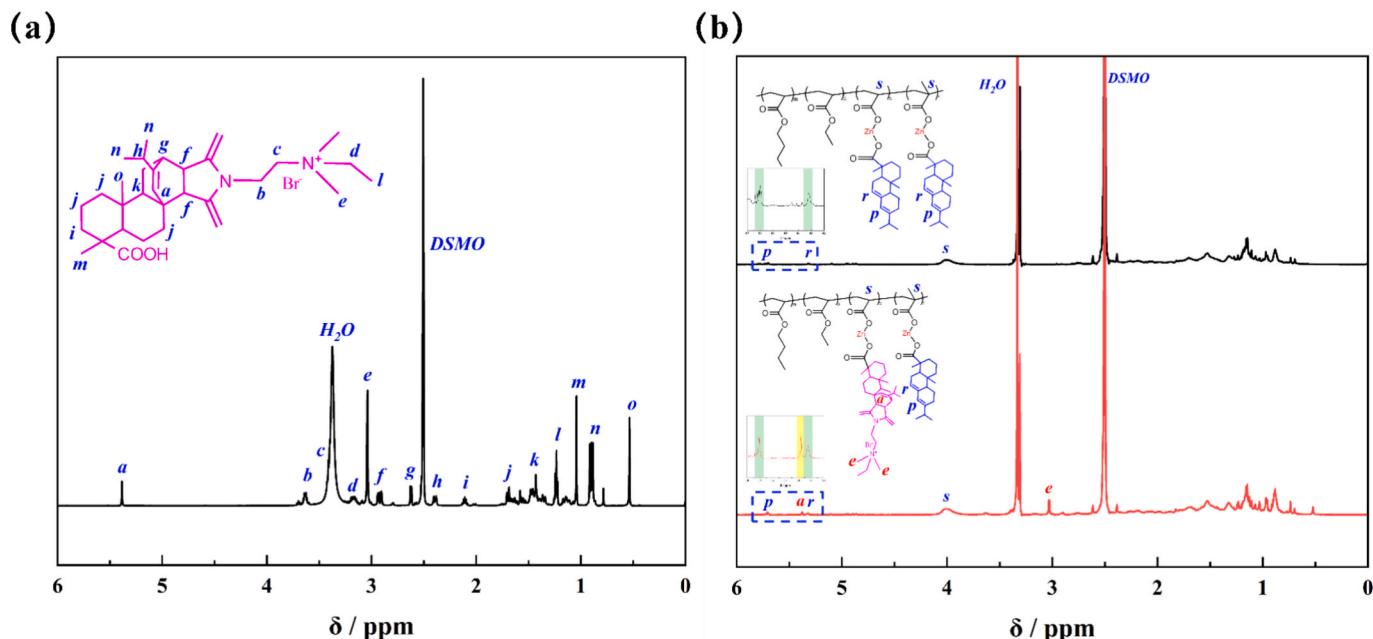


Fig. 2. ^1H NMR spectra of (a) MPA- N^+ and (b) RN-0 and RN-20.

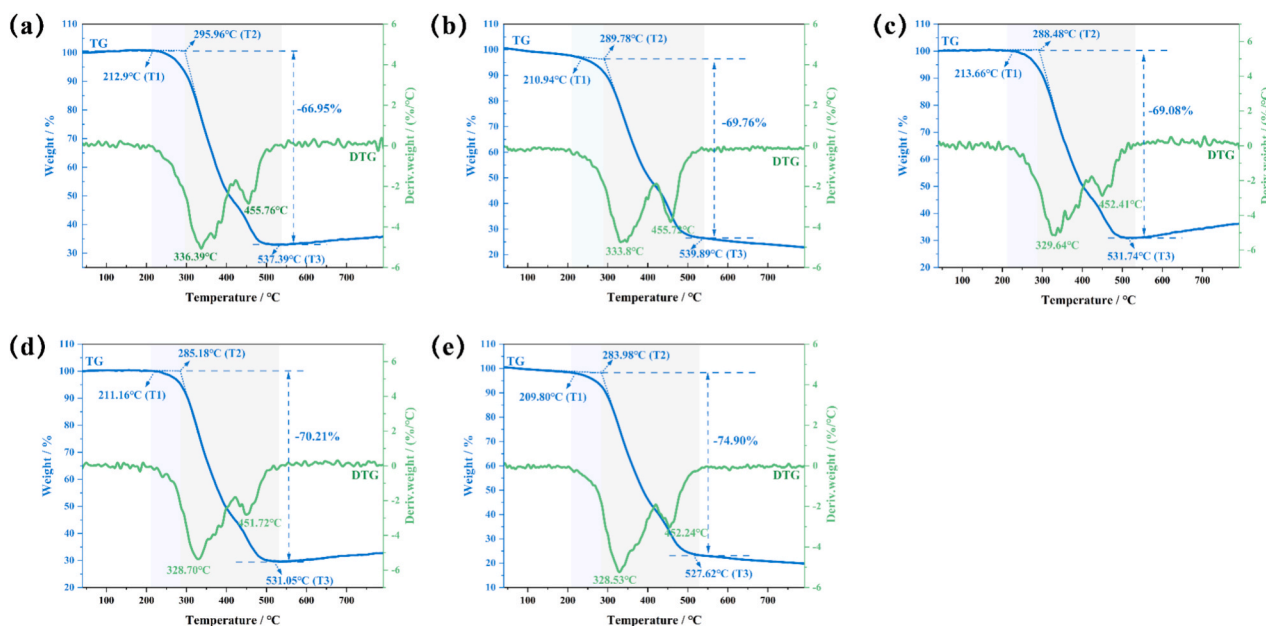


Fig. 3. TG-DTG curves of (a) RN-0, (b) RN-5, (c) RN-10, (d) RN-15 and (e) RN-20.

T1 and T2), which may be due to the loss of moisture and solvents. [20]. The temperature at which mass loss initiates (T1) is comparable for both coatings in this region. Region II (spanning from T2 to T3) represents significant coating mass loss, associated with the decomposition of the coating polymer [34]. Compared to RN-0, RN-20 shows lower T2 (283.95 °C) and T3 (527.62 °C) values than RN-0's T2 (336.39 °C) and T3 (537.39 °C). Additionally, the overall mass loss indicates that RN-20 experiences a mass loss of 74.90 %, higher than RN-0's 66.95 %, suggesting that a larger portion of the RN-20 polymer undergoes pyrolysis. The DTG curve reflects the rate of weight loss, with the peak position indicating the maximum weight loss rate.

The two peaks observed in RN-20 appeared at 328.53 °C and 452.24 °C, both lower than the corresponding peaks of RN-0 at

336.39 °C and 455.76 °C. Similar patterns are evident in the thermogravimetric curves of other RN- x polymers. These results indicate that the RN-20 polymer exhibits lower thermal stability than RN-0, suggesting that the introduction of MPA- N^+ may adversely affect the coating's thermal stability. Nonetheless, all RN- x coatings demonstrated initial pyrolysis temperatures above 200 °C, indicating they still possess good and comparable thermal stability.

The surface valence and elemental composition of the RN- x polymers were further analyzed using XPS (Fig. 4). For RN-20, the spectra confirmed the presence of C, N, O, Zn, and Br, as shown in Fig. 4a. The C 1s peak serves as a reference for the other elements. The atomic contents of C, N, O, Zn, and Br are illustrated in the same figure. In the high-resolution C 1s spectrum (Fig. 4b), the four peaks at 284.8, 286.58,

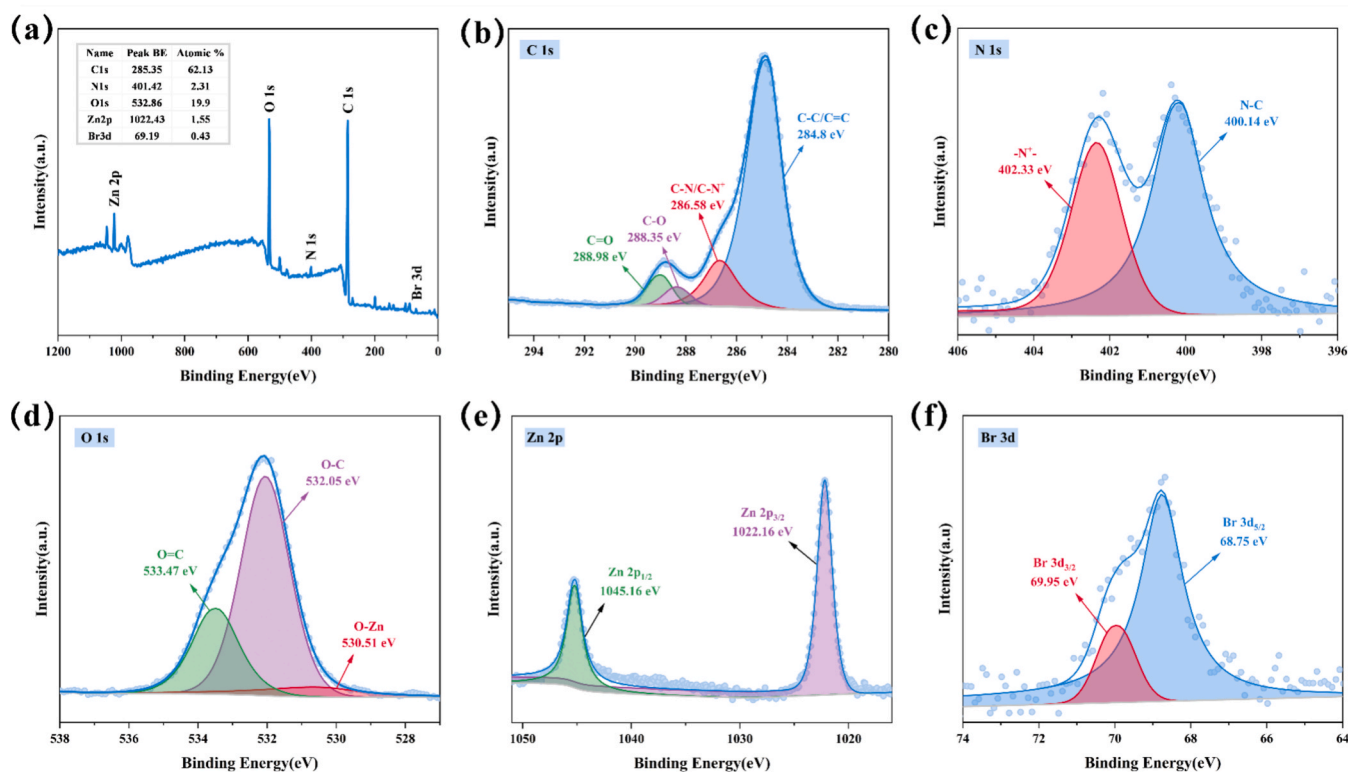


Fig. 4. XPS spectra of RN-20: (a) Survey spectrum, (b) C 1s spectrum, (c) N 1s spectrum, (d) O 1s spectrum, (e) Zn 2p spectrum, and (f) Br 3d spectrum.

288.35, and 288.98 eV correspond to C-C/C=C, C-N/C-N⁺, C-O, and C=O [35]. The presence of the C-N/C-N⁺ bond indicates the existence of MPA-N⁺ on the coating surface. In Fig. 4c, the N 1s spectrum reveals two characteristic peaks: N-C at 400.14 eV and -N⁺ at 402.33 eV, further confirming the successful synthesis and grafting of the MPA-N⁺ molecule [19]. In the O 1s spectrum (Fig. 4d), the peak at 530.51 eV corresponds to O-Zn, while the peak at 532.05 eV is attributed to C-O, and the peak at 533.47 eV corresponds to C=O [36]. The presence of O-Zn confirms the successful grafting of MPA-N⁺ and rosin acid onto the resin side chains. In the Zn 2p spectrum (Fig. 4e), the two peaks at 1022.16 eV and 1045.16 eV correspond to Zn 2p_{3/2} and Zn 2p_{1/2}, respectively [37], confirming the successful synthesis of RN-20. Fig. 4f illustrates the Br 3d spectrum, which contains two peaks at 68.75 eV and 69.95 eV, corresponding to Br 3d_{5/2} and Br 3d_{3/2}, respectively [38]. This result further confirms the successful synthesis of RN-20, which aligns with the analysis of the ¹H NMR spectra.

Coatings must maintain surface stability in practical applications, making it crucial to observe the apparent morphology of coatings before and after immersion in natural seawater. We examined the surface morphology and roughness changes of RN-x coatings before and after static seawater immersion and dynamic seawater simulation experiments using confocal laser scanning microscope (Fig. 5; Fig. S4, Supplementary Material). Before immersion, the coating surface was smooth and flat, devoid of wrinkles and scratches, demonstrating the excellent film-forming properties of the RN-x polymer coating. In the static seawater immersion experiments (RN-0, RN-10, RN-20, panels a-c), the surface of the coatings exhibited slight roughening with a few marks, yet remained intact and free from wrinkles and cracks. In the more intense dynamic seawater simulation experiments (panels d-f), after 35 days of washout, the microscopic surfaces of the coatings became rougher with additional marks, but still maintained integrity, showing no wrinkles or cracks.

The 3D surface scans reveal that the coating surface remains generally flat, exhibiting no significant deformation, indicating its stability. Height line scan plots demonstrate that the microscopic surface

undulation increases with higher concentrations of MPA-N⁺. Both the static seawater immersion and dynamic seawater simulation experiments demonstrated that surface traces and line roughness (R_a) increased with higher concentrations of MPA-N⁺. This phenomenon may be attributed to two factors: first, MPA-N⁺ is a quaternary ammonium molecule with notable hydrophilicity, which can lead to rapid localized hydrolysis of the coating surface [39]; The second factor involves the cation- π interaction between the MPA-N⁺ molecule and the rosin acid molecule. However, in seawater, water molecules interact with and attract the N⁺, leading to a reduction in the strength of the cation- π interaction [40]. This diminished interaction contributes to decreased stability of the coating surface, resulting in increased roughness of the coating.

Hydrophilic and hydrophobic performance tests, along with self-polishing performance evaluations, are crucial for assessing marine antifouling coatings. In this study, we examined the contact angle variation of RN-x coatings. Following the introduction of MPA-N⁺, the water contact angle of the RN-x coating consistently decreased, as shown in Fig. 6a, from 87.4° in the initial RN-0 coating to 73.5° in the RN-20 coating. This is due to the increase of quaternary ammonium groups on the coating surface, which enhances the wettability of the coating [41]. After 35 days of static immersion experiments (Fig. 6b) and dynamic simulation experiments (Fig. 6c), the water contact angle of the coated surface decreased, while the rate of coating mass loss accelerated with increasing MPA-N⁺ content (Fig. 6d, e). This is attributed to the higher MPA-N⁺ concentration on the coating surface, which enhances water wettability and accelerates hydrolysis. The exposed carboxyl groups after hydrolysis are hydrophilic, leading to increased brittleness of the molecular chain and easier detachment from the coating surface, thus further accelerating hydrolysis [34,42,43]. This also means that a higher MPA-N⁺ content in the RN-x coating results in a smaller contact angle before and after hydrolysis. The results indicate that increased MPA-N⁺ content promotes hydrolysis.

Due to the enhanced hydrophilicity from the introduction of MPA-N⁺ into RN-x, we initially had concerns about the coating's adhesion

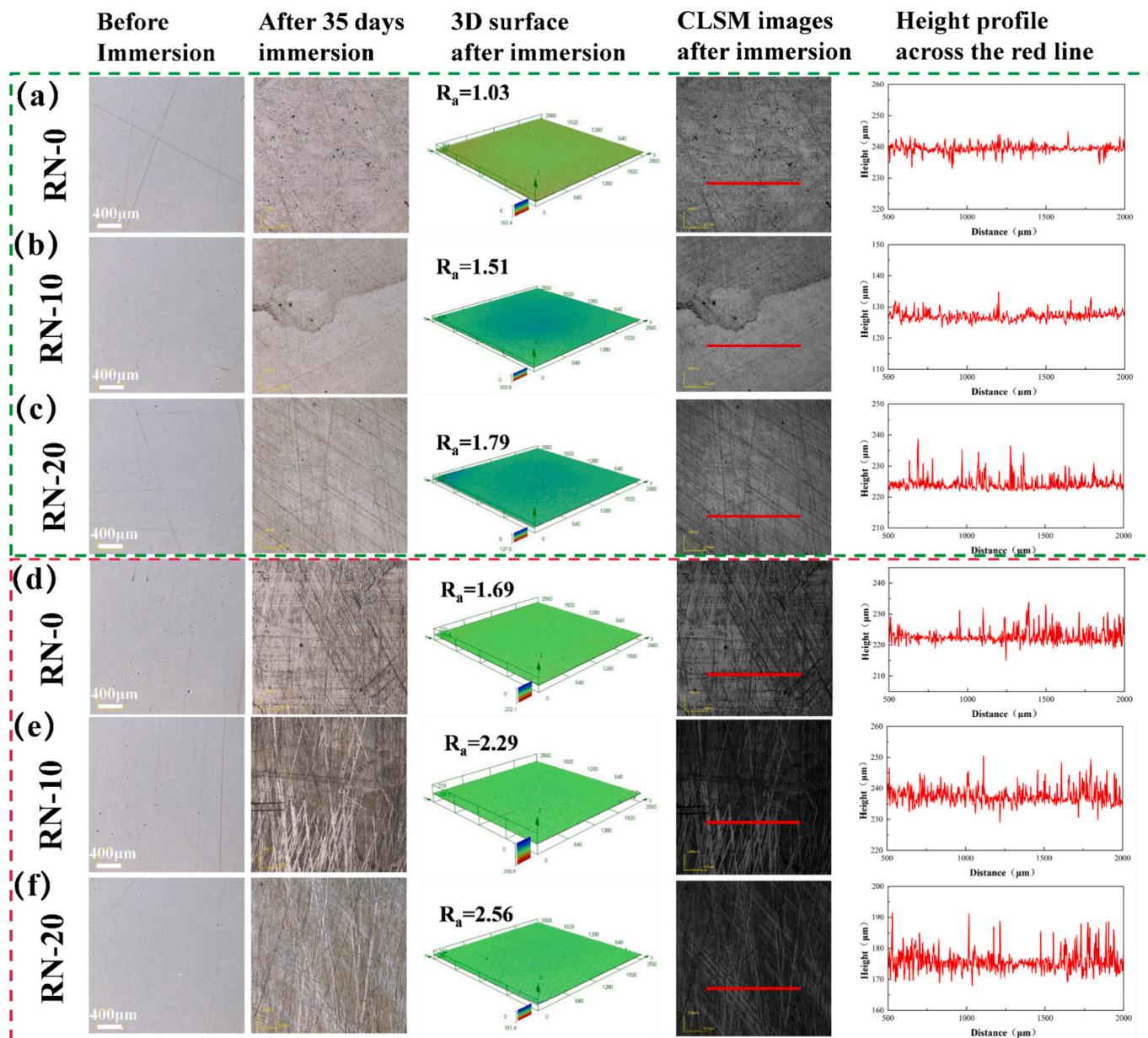


Fig. 5. CLSM profiles of RN-x: (a) RN-0, (b) RN-10, (c) RN-20 coatings after static immersion test and (d) RN-0, (e) RN-10, (f) RN-20 coatings after dynamic immersion test (left to right: surface topography before immersion ($2568 \times 2575 \mu\text{m}^2$) and surface topography ($2568 \times 2575 \mu\text{m}^2$), 3D topography image, 2D topography image and height profile across the red line after 35 days immersion).

performance. At the initial stage of increasing MPA- N^+ quaternary ammonium compound content, the adhesion of the RN-x coating to the epoxy substrate improved. This enhancement may be attributed to the cationic surfactant nature of quaternary ammonium salts, which can reduce the surface tension of the coating system. This reduction facilitates better spreading of the coating on the substrate surface, minimizes defects such as craters and bubbles, improves actual contact between the coating and the substrate, and enhances physical adsorption (e.g., van der Waals forces), thereby increasing adhesion. Additionally, the acrylic resin molecular chains typically contain polar functional groups such as carboxyl ($-\text{COOH}$) and hydroxyl ($-\text{OH}$). The polar head groups of MPA- N^+ quaternary ammonium salts can interact with these groups through hydrogen bonding or ionic interactions, strengthening intermolecular forces within the resin, improving the mechanical integrity of the coating, and reducing its tendency to delaminate from the substrate. However, excessive amounts of quaternary ammonium salts (RN-20) can lead to the formation of locally high-density positive charges within

the coating. The resulting electrostatic repulsion among like charges prevents close packing of the resin molecular chains, thereby reducing adhesion. Nevertheless, all coatings exhibited bond strengths exceeding 4.5 MPa, indicating excellent adhesive performance.

Fig. 6h illustrates the coatings applied to a non-fluorescent paper, curing rapidly at ambient temperature and exhibiting a fluorescent image under 370 nm UV light. It is well known that most aggregation-induced emission (AIE) originates from restriction of intramolecular motion (RIM) [44]. The polymer coatings have the hydrogenated phenanthrene structure, where the spatial site-barrier effect produced by the conjugated structure and the rigid aliphatic ring restricts the molecular motion and hinders the non-radiative leaps, thus enhancing the enhanced photoluminescence properties [45,46]. Fig. 6i presents the fluorescence emission spectrum of the RN-x series coatings. The pre-polymer coating without the addition of rosin or MPA- N^+ showed no significant fluorescence. Among the RN-x coatings, RN-0 exhibited the highest fluorescence emission intensity, while the fluorescence of RN-5,

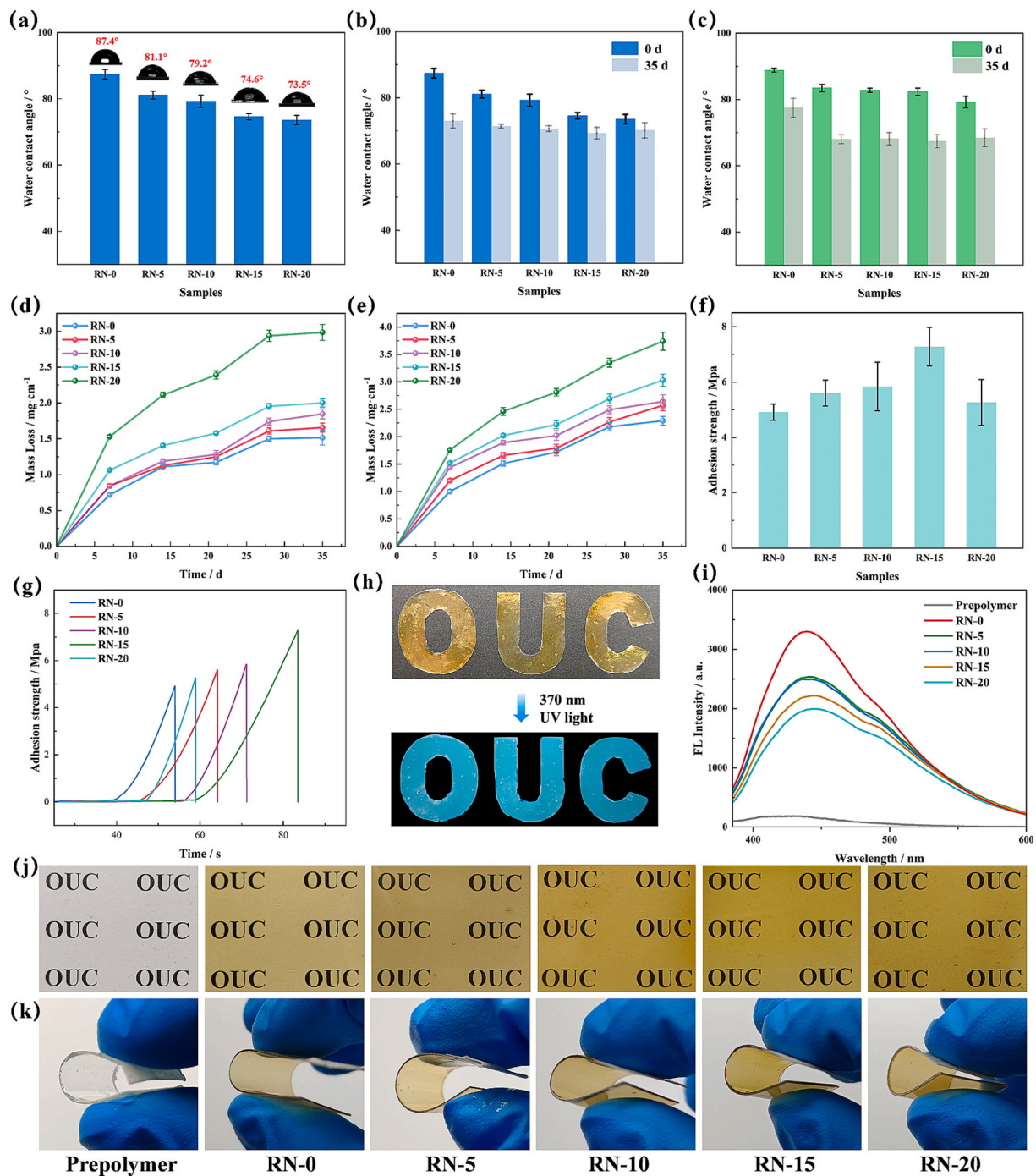


Fig. 6. Water contact angle of (a) RN-x coatings, (b) Variation of angle before and after static immersion experiment, (c) Variation of angle dynamic simulation experiment; Mass loss of RN-x coatings: (d) static immersion experiment, (e) dynamic simulation experiment; (f) Adhesion strength of RN-x on epoxy substrate. (g) Coating Adhesion Test Curves, (h) Fluorescence images of different patterns of RN-20 coating, (i) Fluorescence emission spectrum of RN-x, (j) Transparency of prepolymer and RN-x coatings and (k) Bending test photos of the coatings.

RN-10, RN-15, and RN-20 progressively decreased. This suggests that rosin contributes more significantly to the fluorescence effect than MPA- N^+ and serves as the primary fluorescent component in the system. The fluorescence effect of the coating can effectively assist the coating in resisting the adhesion of algae.

Additionally, the resin coatings demonstrate good transparency and flexibility (Fig. 6j, k). The prepolymer is entirely transparent, while the RN-x coatings display a light brown color due to the presence of rosin, yet they still maintain good transparency. These coatings easily accommodate various bending scenarios and exhibit excellent

flexibility, which is crucial for handling complex applications.

3.2. Antifouling performance test

Proteins can act as a regulatory layer that promotes bacterial attachment and subsequent biofilm growth, marking the initial stage of bioattachment. Therefore, inhibiting protein attachment is crucial for marine coatings. As shown in Fig. 7a, the anti-protein adsorption rate of the RN-0 coating was 57.88 %. In contrast, the anti-protein adsorption capability of RN-x was significantly enhanced after grafting with MPA-

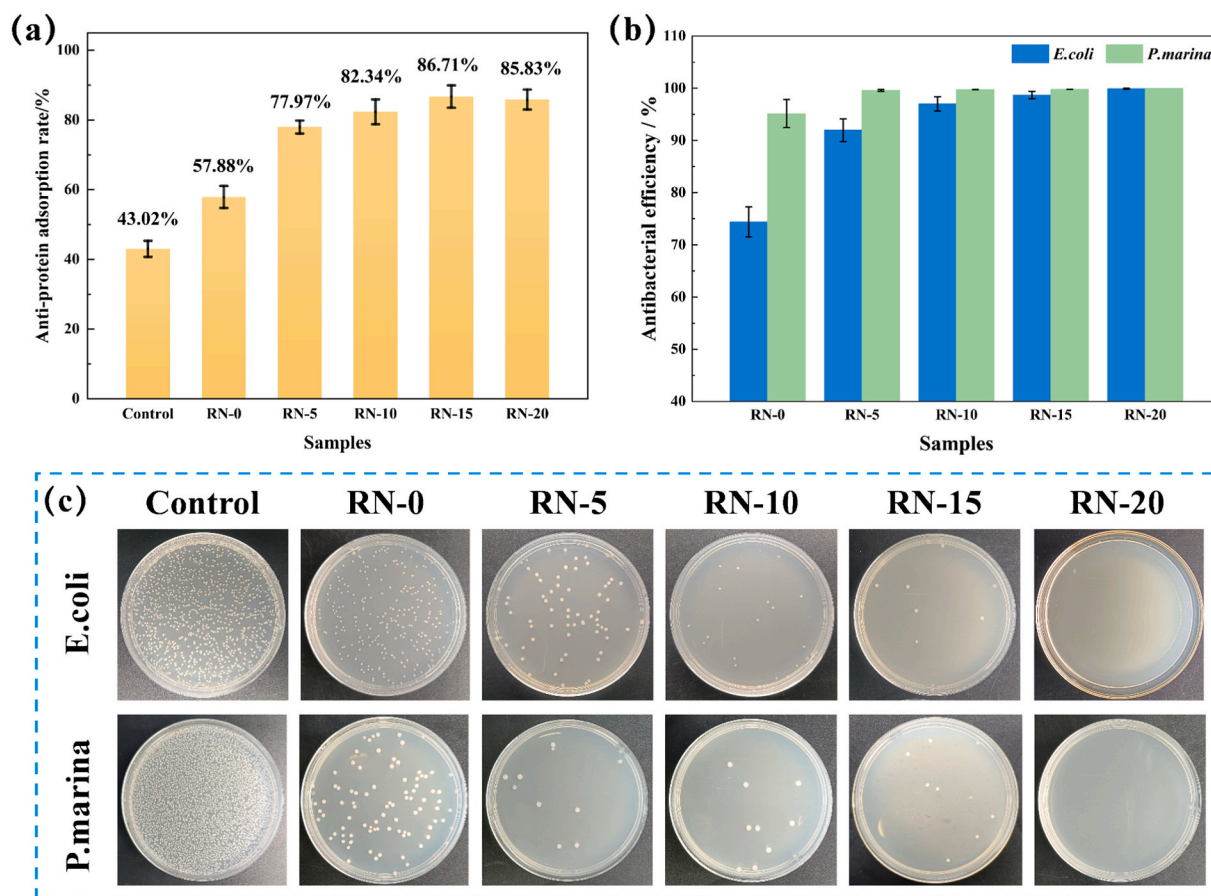


Fig. 7. (a) Protein adsorption resistance of the RN-x coatings; Inhibition graph of RN-x coatings against *E. coli*, *P. marina*, (b) Antibacterial efficiency of RN-x coatings, (c) Photographs of colonies on agar plates.

N^+ quaternary ammonium salt, with RN-15 and RN-20 both exceeding 85 %. This improvement is primarily due to the grafting of MPA- N^+ , which significantly enhances the hydrophilicity of the coating surface. This leads to a dual effect: hydrophilic surfaces are more likely to hydrate, forming repulsive barriers in an aqueous environment, which increases resistance to protein adhesion [47,48]; Secondly, increased hydrophilicity accelerates the hydrolysis of the surface, and the self-renewal process further complicates protein adhesion. This dual action enhances the coating's resistance to biofouling.

Bacterial growth initiates the attachment of marine fouling organisms, making it essential to assess the antibacterial properties of the RN-x coating (Fig. 7b, c). The RN-0 coating demonstrates inherent effectiveness against bacteria, attributable to the synergistic effects of the rosin acid in the coating's side chain and the Zn^{2+} ions released during hydrolysis [49]. The antibacterial efficacy of the coating was significantly enhanced following the grafting of MPA- N^+ . Notably, even in *E. coli*, which was inadequately inhibited by the RN-0 coating, the inhibition rate improved from 74.3 % to 92.0 % upon the introduction of 5 % MPA- N^+ , ultimately exceeding 99.9 % in the RN-20 coating. This enhancement is attributed to the positively charged central structure of MPA- N^+ , whose antimicrobial mechanism is widely understood to involve the adsorption of positively charged quaternary ammonium salt molecules onto the negatively charged surfaces of bacteria via Coulombic forces. Subsequently, the alkyl chains of the quaternary ammonium salts interact hydrophobically with the phospholipid bilayer of cell membranes, leading to their insertion and cleavage within the membranes, resulting in the leakage of intracellular proteins, DNA, RNA, and other contents, ultimately causing bacterial cell death [50–54]. Meanwhile, in this RN-x system, the destruction of bacterial cell membranes by quaternary ammonium molecules facilitates the

entry of Zn^{2+} ions (~ 0.1 mg/g is released by the coating, Table S3) and rosin acid molecules produced by hydrolysis into the bacterial cells. This enhances the synergistic antimicrobial effect, resulting in a more pronounced antimicrobial efficacy.

Inhibition of algal growth and resistance to algal adhesion are crucial properties of marine antifouling coatings. Fig. 8a illustrates the ability of the RN-x coatings to inhibit algal growth. All coatings exhibited effective inhibition (≥ 70 %) of *C. vulgaris*, with the inhibition effect slightly enhanced by increasing the quaternary ammonium salt content. This phenomenon was similarly observed with *P. tricornutum*. The growth inhibition effect was particularly pronounced for *N. closterium*, with inhibition significantly enhanced by increasing the amount of MPA- N^+ quaternary ammonium salt, showing sustained growth inhibition of 85.2 %, up from 45.8 % in the RN-0 coating. Fig. 8b presents the results of experiments assessing the coating's resistance to algal adhesion. The RN-x coatings exhibited notable resistance to algae adhesion, attributed to the coatings' fluorescent repellent effect [27,55]. Even the RN-0 coating, which lacks MPA- N^+ but exhibits strong fluorescence, achieves an algal adhesion resistance rate exceeding 25 %. For *N. closterium*, this resistance rate surpasses 40 %. And as MPA- N^+ content increased, the anti-adhesion ability of the coating improved significantly across all three algal species. This enhancement may be attributed to the increased hydrophilicity of the coating surface resulting from the introduction of quaternary ammonium salts. Previous studies have indicated that algae preferentially adhere to hydrophobic surfaces rather than hydrophilic ones, leading to poor adhesion on hydrophilic surfaces [56]. The hydration layer formed by the surface's hydrophilicity is crucial in preventing microorganisms from making direct contact with the surface, as they must overcome a specific energy barrier to breach this hydration layer [47,57]. Additionally, the hydrolysis of the self-polishing coating

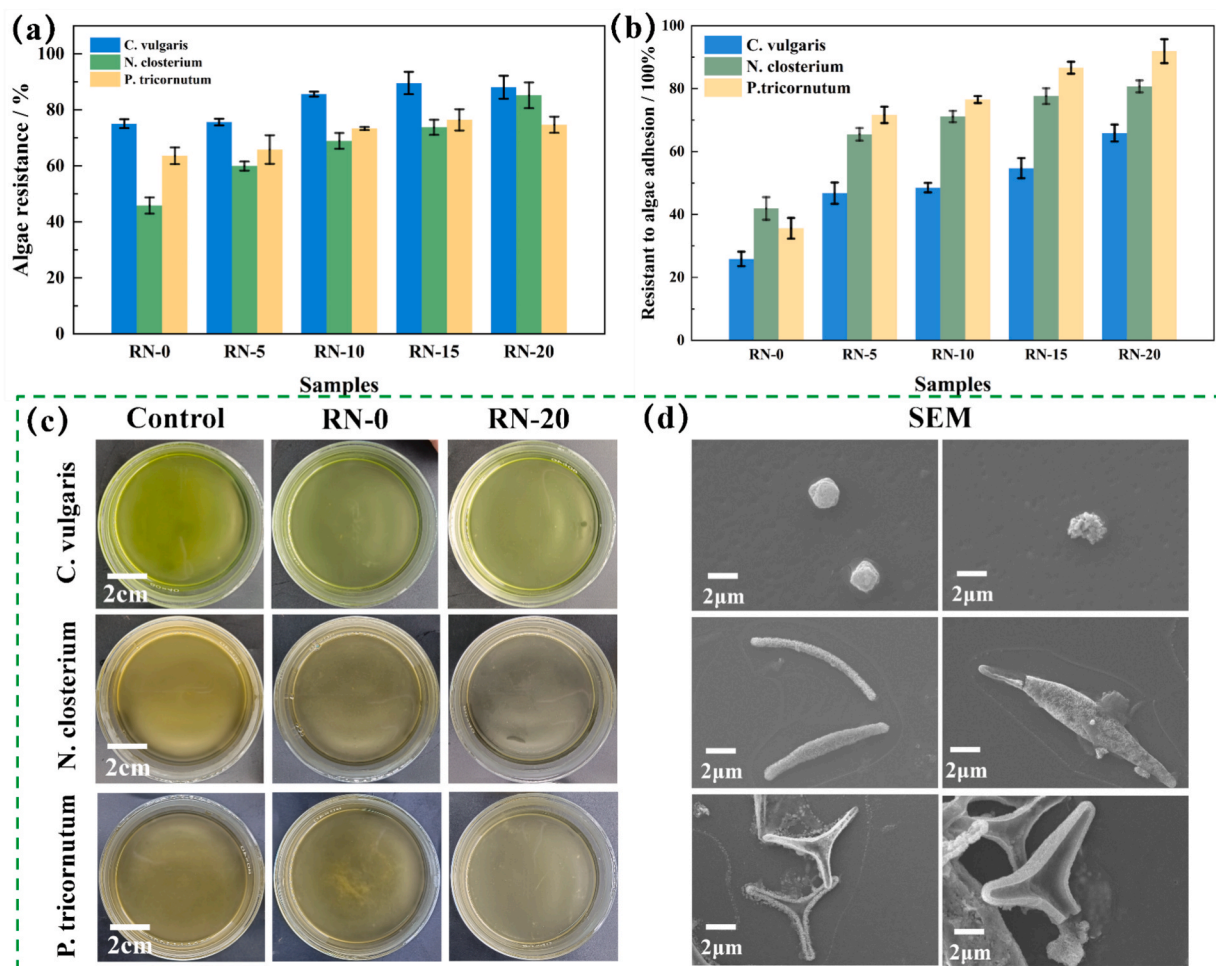


Fig. 8. Resistance to algae and resistance to algae attachment experiments, (a) Algae inhibition rate of RN-x, (b) Resistance to algae adhesion of RN-x, (c) Inhibition of algae growth pictures, (d) SEM images of the adhered algae cells: on the surface of coating RN-0 (left) and coating RN-20 (right).

accelerates the surface renewal rate, further impeding algal attachment.

Fig. 8c illustrates the impact of co-culturing algae with coated sample plates after 7 days, clearly demonstrating that the increase in MPA-N⁺ grafting on the coatings correlates with a decrease in algal concentration, indicating effective inhibition of algal growth. The lipid membranes of algal cells are primarily composed of a phosphate backbone, polysaccharides, and protein couplings, resulting in an overall negative charge [58,59]. MPA-N⁺ possesses a high positive charge density, facilitating electrostatic interactions with microbial lipid membranes [51]. These interactions enable the alkyl chains of MPA-N⁺ to penetrate the microbial cell wall, leading to cellular damage by destabilizing the cytoplasmic membrane. We further verified the mechanism of coating inhibition of algal growth using SEM. As shown in Fig. 8d, the morphology of *C. vulgaris*, *N. closterium* and *P. tricornutum* in contact with the RN-0 surface appeared normal. In contrast, the three algae in contact with the RN-20 coated surface (right side of Fig. 8d) exhibited abnormal shapes and signs of damage. This may be attributed to the action of MPA-N⁺, which disrupts algal cell membranes through a contact-killing mechanism [49]. This leaves the algae defenseless, leading to algal death as a result of the synergistic effects of MPA-N⁺, Zn²⁺, and rosin.

Testing in real marine application scenarios is essential for marine antifouling polymer coatings. In this study, we conducted tests in the port of Qingdao, as illustrated in Fig. 9. The antifouling performance of the RN-x coating was assessed in the waters of Qingdao harbor. After 30 days, significant biofouling was observed on the blank samples, whereas the coated samples remained exceptionally clean. By day 60, substantial

algae and biofilm had accumulated on the blank samples, while silt and biofilm were present on the RN-0 samples. In contrast, the polymer coatings containing MPA-N⁺ (RN-5/10/15/20) exhibited clean surfaces even after 60 days. By 240 days, the blank samples were entirely covered with marine organisms and fouling, while RN-0 displayed significant amounts of silt and biofilm. Conversely, the RN-5/10/15/20 polymer coatings remained largely free of marine organisms, with only minimal silt accumulation, demonstrating excellent antifouling performance. This indicates that incorporating MPA-N⁺ into polymer coatings significantly enhances resistance to marine organisms through the synergistic effects of fluorescent antifouling, contact killing, and self-polishing. This allows these coatings to endure over 240 days of marine fouling resistance in complex marine environments.

4. Conclusion

In conclusion, we have developed a novel self-polishing surface that integrates UV fluorescence-responsive antifouling with contact antifouling, demonstrating exceptional “offensive capabilities” to repel and kill, as well as “defensive capabilities” for surface release. The acrylate prepolymers were synthesized via free radical polymerization, and the RN-x polymers were produced through acid-base reactions using natural rosin and MPA-N⁺ as side chains suspended on the prepolymers. The resin coating effectively repels algae in seawater due to its fluorescent response, which reduces the settlement of marine organisms. The MPA-N⁺ on the polymer surface interacts with negatively charged bacteria through electrostatic forces, allowing the long hydrophobic chains to

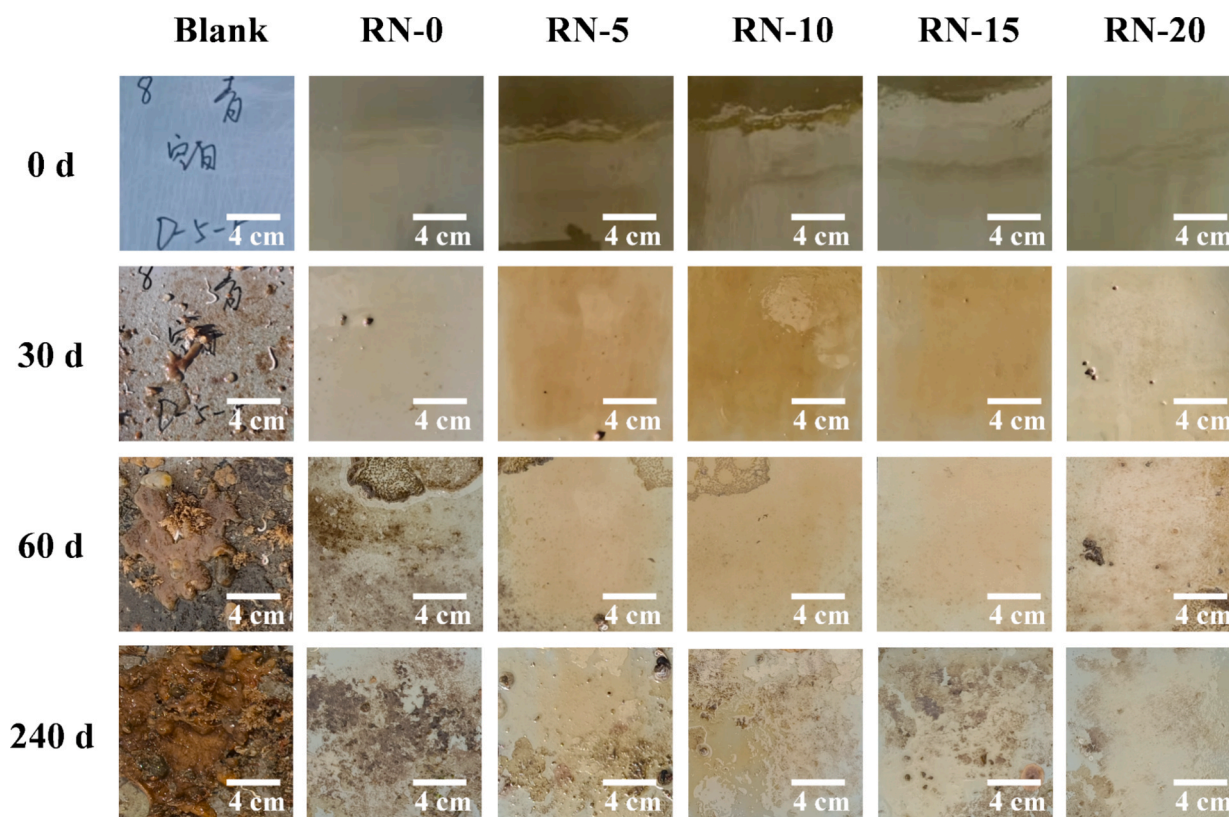


Fig. 9. Photographs of the RN-x coatings and blank immersed in the actual marine environment.

insert into the phospholipid bilayer. This interaction effectively disrupts the cell membranes of bacteria and algae, significantly enhancing the antimicrobial efficacy of the coated surface. Additionally, acrylate polymers undergo hydrolysis and exhibit excellent surface renewal capabilities. This process not only releases Zn^{2+} ions (~ 0.1 mg/g) and natural rosin to enhance sterilization and inhibit algal growth but also facilitates the detachment of fouling materials from the coating surface, allowing for the regeneration of functional surfaces with potent antimicrobial properties. This results in a coated surface that demonstrates exceptional self-renewal and significant antifouling capabilities, encompassing resistance to protein adhesion, antimicrobial properties, inhibition of algal growth, and resistance to algal adhesion. This polymer coating, characterized by its environmentally friendly synergistic antifouling mechanism, offers a novel strategy for the development of sustainable long-term antifouling coatings.

CRedit authorship contribution statement

Hao Xu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Guangling He:** Writing – review & editing, Methodology. **Chunhua Ni:** Writing – review & editing, Formal analysis. **Xia Li:** Validation, Supervision, Conceptualization. **Xiaohui Jiang:** Resources, Project administration. **Liangmin Yu:** Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsusc.2025.163861>.

Data availability

The authors do not have permission to share data.

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